# Practical No. 5

**Concurrency, Media and Security**

1. Develop an Android application demonstrating Threads and Handlers for background processing.
2. Develop an Android application implementing Media APIs and Telephone APIs.
3. Develop an Android application implementing Runtime Permissions and security handling in AndroidManifest.xml.

## 1. Develop an Android application demonstrating Threads and Handlers for background processing.

**MainActivity.java**

```
package com.example.myapplication;
import android.annotation.SuppressLint;
import android.os.Bundle;
import android.os.Handler;
import android.os.Message;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
   TextView textView;
   Handler handler;
   @SuppressLint("MissingInflatedId")
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      textView = findViewById(R.id.textView);
      handler = new Handler() {
         @Override
         public void handleMessage(Message msg) {
            textView.setText("Count : " + msg.arg1);
         }
      };
      Thread thread = new Thread(new Runnable() {
         @Override
         public void run() {
            for(int i=1;i<=10;i++){
               Message message = new Message();
               message.arg1 = i;
               handler.sendMessage(message);
               try {
                  Thread.sleep(1000);
```

```
                } catch (InterruptedException e) {
                    e.printStackTrace();
                }
            }
        }


    });
    thread.start();  }
}
```

**activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:gravity="center"
   android:orientation="vertical">
   <TextView
      android:id="@+id/textView"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Thread Example"
      android:textSize="25sp"/>
</LinearLayout>
```


My Application
Count : 10